

ASHRITHA ABBU

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with hands-on experience delivering end-to-end machine learning pipelines, Retrieval-Augmented Generation systems, transformer-based NLP models, and computer vision solutions. Proficient in Python, PyTorch, TensorFlow, and LLM fine-tuning. Adapt at translating complex model outputs into measurable business impact. Seeking an AI/ML or Software Engineering role to architect intelligent, data-driven systems at scale.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML/CSS

ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers

Generative AI & LLMs: RAG Pipelines, Fine-Tuning, Prompt Engineering, Prompt Tuning, Transfer Learning

Automation & APIs: n8n Workflow Automation, FastAPI, REST APIs

Data & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Feature Engineering, EDA

Tools & Platforms: Git, GitHub, Jupyter Notebook, Streamlit, VS Code, SQL Server Management Studio

PROFESSIONAL EXPERIENCE

AI Engineer Intern | Data Science & AI Internship

Oct 2025 – Present

- Architected and deployed production-grade RAG pipelines integrating vector stores and LLMs, enabling accurate context-aware Q&A over domain-specific document corpora reducing manual document retrieval time significantly.
- Engineered prompt tuning strategies to steer LLM behavior without full fine-tuning, improving output quality across multiple task types and reducing inference costs.
- Built end-to-end ML workflows (data collection, preprocessing, feature engineering, model training, evaluation) using Pandas, NumPy, and Scikit-learn, accelerating model iteration cycles.
- Developed interactive dashboards and visualizations with Matplotlib and Seaborn, translating complex model outputs into actionable insights for non-technical stakeholders.

AI/ML Virtual Intern | EduSkills Foundation

Jul 2024 – Sep 2024

- Completed a structured AI/ML curriculum covering Python, supervised/unsupervised learning algorithms, and end-to-end ML workflow design, delivering a certified capstone project.

PROJECTS

NewsGPT — Domain-Specific Language Model — Python · PyTorch · Transformers · NLP

- Built a GPT-style transformer from scratch on a multi-category news corpus (sports, politics, technology), implementing multi-head self-attention, token/positional embeddings, output projection heads using PyTorch.
- Automated a data-ingestion pipeline to fetch, clean, and balance news articles from diverse sources, ensuring high-quality training data across all categories.
- Monitored training and validation loss curves on GPU-accelerated infrastructure, iterating on architecture to reduce overfitting and improve generalization.

Forest Fire Detection — Computer Vision System — Python · TensorFlow · Transfer Learning · Streamlit

- Fine-tuned a pre-trained CNN via transfer learning for binary fire/non-fire image classification, achieving high accuracy through targeted data augmentation and preprocessing techniques.
- Deployed the model as a Streamlit web application, enabling real-time image upload and instant fire-detection inference to support early-warning and safety use cases.

EDUCATION

B.Tech in Computer Science & Engineering

Nov 2021 – Jun 2025

Kamala Institute of Technology and Science · CGPA: 7.4 / 10

Board of Intermediate Education (Class XII)

Jun 2019 – Mar 2021

Sree Chaitanya Junior College · CGPA: 9.6 / 10

CERTIFICATIONS

- AWS Academy Graduate — Cloud Foundations (Amazon Web Services)
- AI Engineer Internship Certificate — EduSkills Foundation
- Python Full Stack Workshop — BrainoVision Solutions Pvt. Ltd.